

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
Windows PowerShell
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PS C:\Users\ritwik> wsl
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [341 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [432 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.7 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [150 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [105 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.0 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [223 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [72.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Fetched 2292 kB in 3s (902 kB/s)
93 packages can be upgraded. Run 'apt list --upgradable' to see them.
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik$ sudo apt install strace
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 93
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [670 kB]
Fetched 737 kB in 3s (288 kB/s)
Selecting previously unselected package libunwind8:amd64.
```

```
(Reading database ... 42277 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik$ cd osp
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP$ cd exec
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ nano program2.c
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ gcc program2.c -o program2
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ ./program2
Cannot open source file: No such file or directory
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ echo "This is sample text for testing the file copy program." > sample.txt
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ nano sample.txt
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ ls -l
total 40
-rwxrwxrwx 1 ritwik ritwik 1292 Aug  5 04:06 program2.c
-rwxrwxrwx 1 ritwik ritwik 16472 Aug  5 03:22 program1
-rwxrwxrwx 1 ritwik ritwik 1056 Aug  5 03:22 program1.c
-rwxrwxrwx 1 ritwik ritwik 16216 Aug  5 04:06 program2
-rwxrwxrwx 1 ritwik ritwik 959 Aug  5 03:35 program2.c
-rwxrwxrwx 1 ritwik ritwik 55 Aug  5 04:10 sample.txt
This is sample text for testing the file copy program.
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ ./program2 sample.txt copy.txt
Cannot open source file: No such file or directory
ritwik@DESKTOP-PK3LAMI:/mnt/c/Users/ritwik/OSP/exec$ strace -o trace_output.txt ./program2 sample.txt copy.txt
cat trace_output.txt
Cannot open source file: No such file or directory
execve("./program2", ["/program2", "sample.txt", "copy.txt"], 0x7fffd4ff6850 /* 26 vars */) = 0
brk(NULL)                               = 0x5fe66ef3b000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x748d34793000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x748d3478e000
close(3)                                = 0
```



```

fcntl(2, F_GETFD)
=
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([fd=0, events=0], [fd=1, events=0], [fd=2, events=0], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x73a60b645cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "6300e94f5000-6300e9c2000 r--p 0", ...1024) = 1024
read(3, "0 00:30 13647", ...1024) = 1024
read(3, "00 00:30 13650", ...1024) = 1024
read(3, " /usr/lib/x86_64-linux", ...1024) = 1024
read(3, "0 13644", ...1024) = 1024
close(3) = 0
sched_getaffinity(1102, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x73a60bb2cb00
mprotect(0x73a60bb2cb00, 4096, PROT_NONE) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
gettid(0) = 1102
rt_sigaction(SIGSEGV, {sa_handler=0x6300e94f5000, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x73a60b645cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=0x6300e94f5000, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x73a60b645cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7fed3be92400, 0x2000, 0, 0) = 4408
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/ucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0444, stx_size=4080, ...}) = 0
read(3, "## Common strings shared across a", ...4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x73a60b645cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x73a60b645cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x73a60b645cb0}, {sa_handler=0x6300e94f5000, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x73a60b645cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x73a60b645cb0}, {sa_handler=0x6300e94f5000, sa_mask=[], sa_flags=SA_RESTORER|SA_SIGINFO, sa_restorer=0x73a60b645cb0}, 8) = 0
fcntl(2, F_GETGID, c_iflag=ICRNL|IXON, c_oflag=NL0|CRNL|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=BS38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IXEXTN|ECHOCTL|ECHOKE, ...1) = 0

```